Dean McPhee

Full-stack Software Engineer

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Full-stack Software Engineer with 3.5+ years' professional experience building and maintaining production systems using technologies like React, Java (Spring Boot), C#, and AWS. I have a proven track record of designing accessible, user-friendly frontends; developing robust, RESTful back-ends; and building scalable cloud infrastructure.

Beyond hands-on development, I've led the delivery of numerous successful features by working closely with product owners to assess feasibility, estimate effort, shape user experience, and define implementation approaches. I've also supported graduate and junior engineers through task delegation, code review, and hands-on mentoring, helping teams deliver high-quality software reliably and efficiently.

I have experience with defence-adjacent projects and currently hold NV1 Security Clearance.

SKILL SUMMARY

Back-end	Java · Spring Boot · Unit Testing (JUnit) · C# · Swagger · SQL · NoSQL · REST APIs
DevOps	AWS EC2 · Lambda · EKS · CloudFormation · CloudWatch · CI/CD (Jenkins) · Linux
Front-end	React.js · Redux.js · JS · CSS · HTML · Responsive design · Web Accessibility (WCAG)
General	Scrum · User Stories · Effort Estimation · Code Review · QA · Incident Response · Git

PROFESSIONAL EXPERIENCE

JAN 2024 - PRESENT
Software Engineer
Leidos Australia – IT, defence & intelligence

- Developed and maintained a complex full-stack web application using modern tech, integrating third-party services and a non-relational database.
- Took full technical ownership of various new features: evaluating potential solutions, estimating implementation effort, and writing user stories to ensure successful on-time delivery.
- Provided technical leadership for a 4-person sub-team, consistently delivering features on schedule with high satisfaction from product owners and end users.
- Implemented and maintained AWS cloud infrastructure components to support deployment and scalability of microservices.
- Debugged critical issues in staging and production environments: documenting incidents, developing permanent fixes, meeting SLAs and minimizing service disruption.
- Identified and fixed performance bottlenecks in legacy code, with improvements as high as 90% for certain key features, greatly improving responsiveness.
- Built and introduced a unit testing framework that lowered regression rates, improved code quality, and raised confidence in releases.

JUL 2022 - DEC 2023
Graduate Software Engineer
Leidos Australia – IT, defence & intelligence

NOV 2020 - JUNE 2022
Graduate Mechatronics Engineer
SPEE3D – Industrial metal-additive design

- Defined and validated tight tolerance stacks for precision assemblies, reducing mechanical runout by 45% while working directly with manufacturers to balance manufacturability, performance, and cost.

- Conducted mechanical and thermal FEA on a mission-critical assembly, producing technical documentation used to achieve certification with an international body.
- Played a key role in migrating to a new PDM system, writing automation scripts that increased productivity across the engineering department.
- Supported and guided newer engineers by sharing knowledge on design tools, analysis methods, and company engineering processes.

SEP 2020 - NOV 2020 **Manufacturing Engineering Intern**
SCHOTT MINIFAB – Microfluidic device manufacturing

- Developed, tested and deployed PLC code to a high-volume production line, adding data acquisition capabilities for improved analysis and debugging.
- Researched and documented improvements to functional testing approaches for mass-produced products, presenting findings to engineering stakeholders.

EDUCATION

MAR 2020 - JUNE 2022 **Master of Engineering (with Distinction), Mechatronics Systems**
The University of Melbourne

- Dean's Honours List (3.9 GPA).
- Relevant coursework: Software Development · Object-Oriented Programming · Algorithms & Data Structures · Numerical Programming · Internet Technologies

MAR 2017 - NOV 2019 **Bachelor of Science (with Distinction), Mechatronics Systems**
The University of Melbourne

- Dean's Honours List (3.8 GPA).

PROJECTS

AUG 2021 - JUN 2022 **Amphibious UAV (Industry-Sponsored)**
The University of Melbourne

- Led the software system design for a team of 4 developing a proof-of-concept amphibious UAV for a defence contractor, capable of covertly transitioning between aerial and submerged operation.
- Built and integrated the autonomy and control stack, managing sensor inputs, actuator outputs, and mission sequencing for precise depth control and reliable operation.
- Achieved the 'Telstra Excellence in Learning and Application' industry-sponsored award at the biannual Melbourne University Endeavour Exhibition.

AUG 2021 - NOV 2021 **Autonomous Warehouse Robot**
The University of Melbourne

- Worked in a multidisciplinary team to develop the full software stack for an autonomous wheeled robot operating in a mock warehouse environment.
- Implemented graph-based path planning (Voronoi roadmap + Dijkstra's algorithm) and closed-loop motion control logic. Integrated encoder-based odometry for real-time pose estimation.
- Designed a modular, node-based system architecture to coordinate planning, control, and actuator subsystems. This allowed for reliable autonomous navigation and task execution.
- Achieved an H1.